IE307 – Công nghệ lập trình đa nền tảng cho ứng dụng di động

BÀI TẬP THỰC HÀNH 2:
CUSTOM COMPONENTS, REACT NAVIGATION

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Lớp: IE307.R11.1

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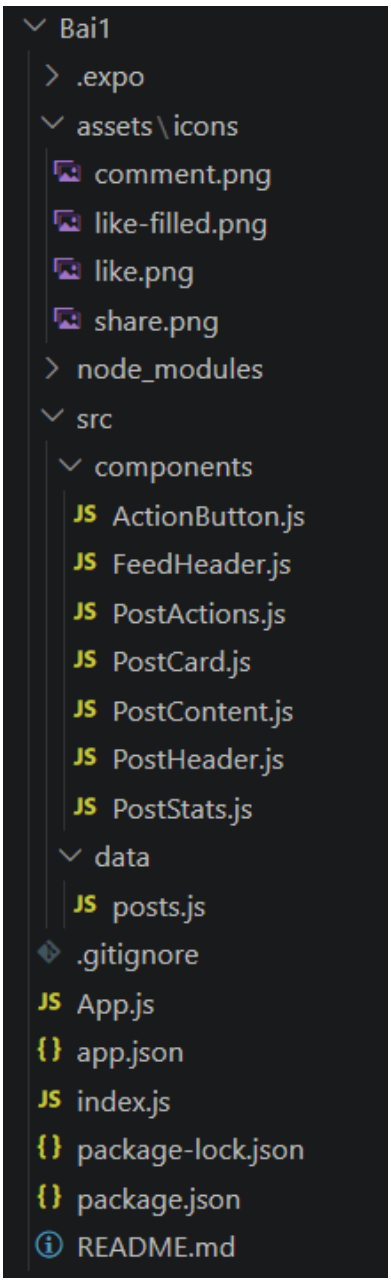
Bài làm

Bài 1: Social Media Feed (cont.)

Link video

https://drive.google.com/file/d/10hqtNb-Zo_mhHegSU8iARM1GCIF395kR/view?usp=drive_link

Cấu trúc project



Các file code

App.js

Bai1 > JS App.js > App

```
// 23521720 - Nguyen Minh Tuan
import { useState } from 'react';
import { FlatList, StyleSheet, View } from 'react-native';
import { SafeAreaProvider, SafeAreaView } from 'react-native-safe-area-context';
import { StatusBar } from 'expo-status-bar';

import FeedHeader from './src/components/FeedHeader';
import PostCard from './src/components/PostCard';
import initialPosts from './src/data/posts';

export default function App() {
  const [posts, setPosts] = useState(initialPosts);

  const updatePost = (postId, update) => {
    setPosts((currentPosts) =>
      currentPosts.map((post) => (post.id === postId ? update(post) : post)),
    );
  };

  const handleLike = (postId) => {
    updatePost(postId, (post) => ({
      ...post,
      liked: !post.liked,
      likes: Math.max(0, post.likes + (post.liked ? -1 : 1)),
    }));
  };

  const handleComment = (postId) => {
    updatePost(postId, (post) => ({ ...post, comments: post.comments + 1 }));
  };

  const handleShare = (postId) => {
    updatePost(postId, (post) => ({ ...post, shares: post.shares + 1 }));
  };

  return (
    <SafeAreaProvider>
      <View style={styles.container}>
        <StatusBar style="light" />
        <FeedHeader />
        <SafeAreaView edges={['left', 'right', 'bottom']} style={styles.feedArea}>
          <FlatList
            data={posts}
            renderItem={({ item }) => (
              <PostCard
                post={item}
                onLike={() => handleLike(item.id)}
                onComment={() => handleComment(item.id)}
                onShare={() => handleShare(item.id)}
              />
            )}
            keyExtractor={(item) => item.id}
            contentContainerStyle={styles.feedContent}
            showsVerticalScrollIndicator={false}
```

```
            />
          </SafeAreaView>
        </View>
      </SafeAreaProvider>
    );
  }

  const styles = StyleSheet.create({
    container: {
      flex: 1,
      backgroundColor: '#eef2f8',
    },
    feedArea: {
      flex: 1,
    },
    feedContent: {
      padding: 12,
      paddingBottom: 28,
    },
  });

```

PostCard.js

Bai1 > src > components > JS PostCard.js > ...

```
// 23521720 - Nguyen Minh Tuan
import { StyleSheet, View } from 'react-native';

import PostActions from './PostActions';
import PostContent from './PostContent';
import PostHeader from './PostHeader';
import PostStats from './PostStats';

export default function PostCard({ post, onLike, onComment, onShare }) {
  return (
    <View style={styles.card}>
      <PostHeader username={post.username} avatar={post.avatar} />
      <PostContent content={post.content} image={post.image} />
      <PostStats likes={post.likes} comments={post.comments} shares={post.shares} />
      <View style={styles.divider} />
      <PostActions liked={post.liked} onLike={onLike} onComment={onComment} onShare={onShare} />
    </View>
  );
}

const styles = StyleSheet.create({
  card: {
    marginBottom: 14,
    backgroundColor: '#ffffff',
    borderRadius: 14,
    shadowColor: '#14213d',
    shadowOffset: { width: 0, height: 3 },
    shadowOpacity: 0.1,
    shadowRadius: 8,
    elevation: 3,
    overflow: 'hidden',
  },
  divider: {
    height: StyleSheet.hairlineWidth,
    marginHorizontal: 14,
    backgroundColor: '#ccd3df',
  },
});

```

PostHeader.js

Bai1 > src > components > JS PostHeader.js > ...

```
// 23521720 - Nguyen Minh Tuan
import { Image, StyleSheet, Text, View } from 'react-native';

export default function PostHeader({ username, avatar }) {
  return (
    <View style={styles.authorRow}>
      <Image source={{ uri: avatar }} style={styles.avatar} />
      <View style={styles.authorText}>
        <Text style={styles.username}>{username}</Text>
        <Text style={styles.postTime}>Just now</Text>
      </View>
    </View>
  );
}

const styles = StyleSheet.create({
  authorRow: {
    flexDirection: 'row',
    alignItems: 'center',
    paddingHorizontal: 14,
    paddingTop: 14,
  },
  avatar: {
    width: 46,
    height: 46,
    borderRadius: 23,
    backgroundColor: '#dfe7f3',
  },
  authorText: {
    flex: 1,
    marginLeft: 11,
  },
  username: {
    color: '#172033',
    fontSize: 16,
    fontWeight: '700',
  },
  postTime: {
    marginTop: 2,
    color: '#778198',
    fontSize: 12,
  },
});

```

PostContent.js

Bai1 > src > components > JS PostContent.js > ...

```
// 23521720 - Nguyen Minh Tuan
import { Image, StyleSheet, Text, View } from 'react-native';

export default function PostContent({ content, image }) {
  return (
    <>
      <Text style={styles.content}>{content}</Text>
      <View style={styles.postImageFrame}>
        <Image source={{ uri: image }} style={styles.postImage} />
      </View>
    </>
  );
}

const styles = StyleSheet.create({
  content: {
    paddingHorizontal: 14,
    paddingVertical: 12,
    color: '#30394c',
    fontSize: 15,
    lineHeight: 21,
  },
  postImageFrame: {
    marginHorizontal: 14,
    borderRadius: 10,
    overflow: 'hidden',
    backgroundColor: '#dfe7f3',
  },
  postImage: {
    width: '100%',
    height: 230,
    borderRadius: 10,
    resizeMode: 'cover',
  },
});
```

PostStats.js

Bai1 > src > components > JS PostStats.js > ...

```
// 23521720 - Nguyen Minh Tuan
import { StyleSheet, Text, View } from 'react-native';

export default function PostStats({ likes, comments, shares }) {
  return (
    <View style={styles.statisticsRow}>
      <Text style={styles.statisticsText}>{likes} Likes</Text>
      <Text style={styles.statisticsText}>{comments} Comments</Text>
      <Text style={styles.statisticsText}>{shares} Shares</Text>
    </View>
  );
}

const styles = StyleSheet.create({
  statisticsRow: {
    flexDirection: 'row',
    justifyContent: 'space-between',
    paddingHorizontal: 14,
    paddingVertical: 11,
  },
  statisticsText: {
    color: '#667085',
    fontSize: 12,
    fontWeight: '600',
  },
});

```

PostActions.js

Bai1 > src > components > JS PostActions.js > ...

```
// 23521720 - Nguyen Minh Tuan
import { StyleSheet, View } from 'react-native';

import ActionButton from './ActionButton';

const actionIcons = {
  like: require('../../assets/icons/like.png'),
  liked: require('../../assets/icons/like-filled.png'),
  comment: require('../../assets/icons/comment.png'),
  share: require('../../assets/icons/share.png'),
};

export default function PostActions({ liked, onLike, onComment, onShare }) {
  return (
    <View style={styles.actionsRow}>
      <ActionButton
        icon={liked ? actionIcons.liked : actionIcons.like}
        label={liked ? 'Liked' : 'Likes'}
        active={liked}
        accessibilityLabel={liked ? 'Unlike post' : 'Like post'}
        onPress={onLike}
      />
      <ActionButton icon={actionIcons.comment} label="Comments" accessibilityLabel="Add a comment" onPress={onComment} />
      <ActionButton icon={actionIcons.share} label="Shares" accessibilityLabel="Share post" onPress={onShare} />
    </View>
  );
}

const styles = StyleSheet.create({
  actionsRow: {
    flexDirection: 'row',
    paddingHorizontal: 8,
    paddingVertical: 7,
  },
});

```

FeedHeader.js

Bai1 > src > components > JS FeedHeader.js > ...

```
// 23521720 - Nguyen Minh Tuan
import { StyleSheet, Text, View } from 'react-native';
import { SafeAreaView } from 'react-native-safe-area-context';

export default function FeedHeader() {
  return (
    <SafeAreaView edges={['top', 'left', 'right']} style={styles.safeArea}>
      <View style={styles.header}>
        <Text style={styles.title}>Social Media Feed</Text>
        <Text style={styles.subtitle}>Connect, share, and enjoy</Text>
      </View>
    </SafeAreaView>
  );
}

const styles = StyleSheet.create({
  safeArea: {
    backgroundColor: '#1264d7',
  },
  header: {
    paddingHorizontal: 20,
    paddingTop: 13,
    paddingBottom: 15,
  },
  title: {
    color: '#ffffff',
    fontSize: 24,
    fontWeight: '800',
  },
  subtitle: {
    marginTop: 3,
    color: '#dce9ff',
    fontSize: 13,
  },
});

```

ActionButton.js

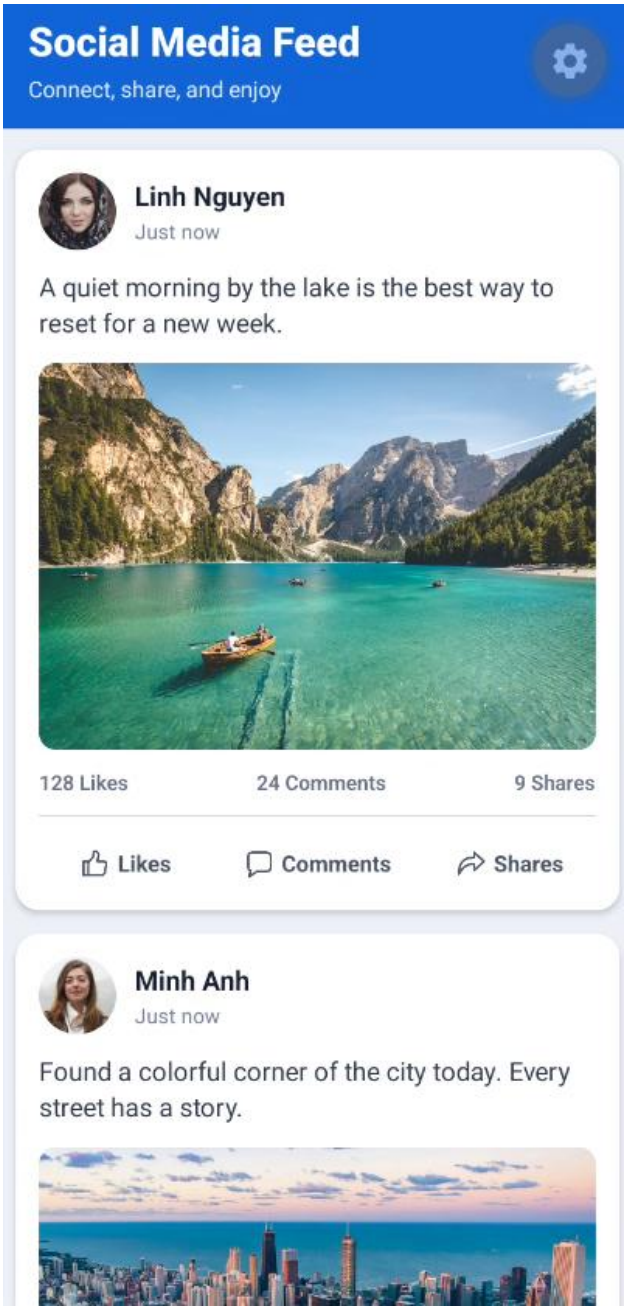
Bai1 > src > components > JS ActionButton.js > ...

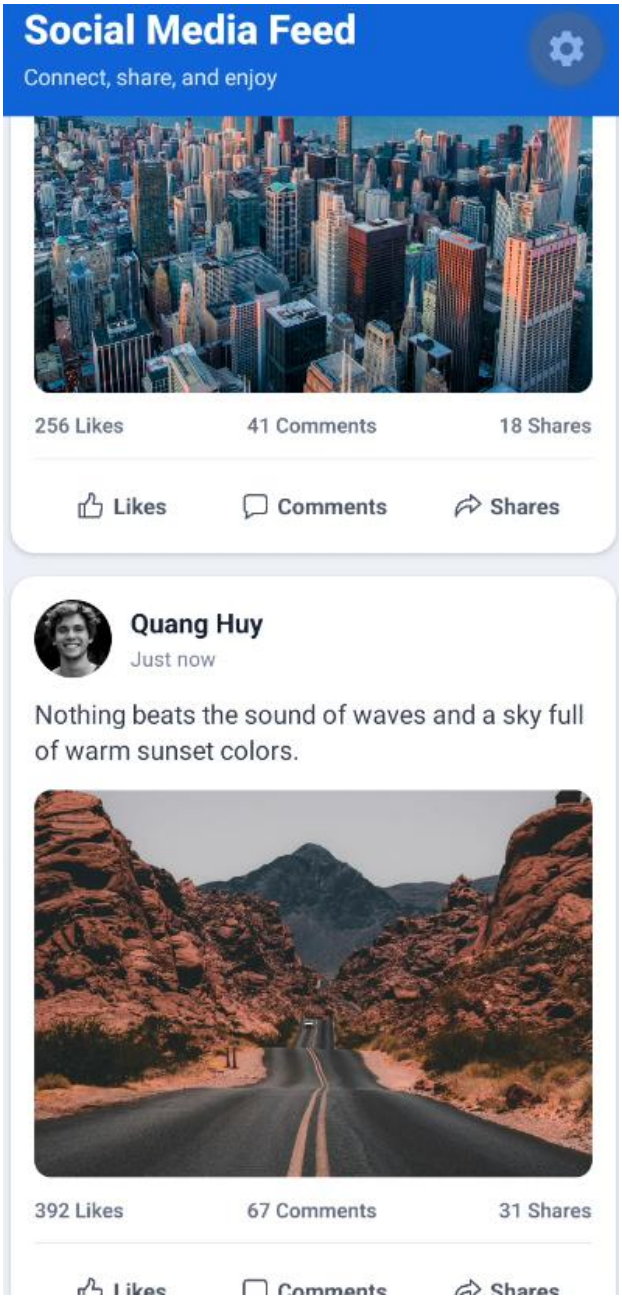
```
// 23521720 - Nguyen Minh Tuan
import { Image, Pressable, StyleSheet, Text } from 'react-native';

export default function ActionButton({ icon, label, active = false, accessibilityLabel, onPress }) {
  return (
    <Pressable
      accessibilityRole="button"
      accessibilityLabel={accessibilityLabel}
      accessibilityState={{ selected: active }}
      onPress={onPress}
      style={({ pressed }) => [
        styles.actionButton,
        active && styles.activeActionButton,
        pressed && styles.pressedActionButton,
      ]}
    >
      <Image source={icon} accessible={false} style={[styles.actionIcon, active && styles.activeActionIcon]} />
      <Text style={[styles.actionText, active && styles.activeActionText]}>{label}</Text>
    </Pressable>
  );
}

const styles = StyleSheet.create({
  actionButton: {
    flex: 1,
    flexDirection: 'row',
    alignItems: 'center',
    justifyContent: 'center',
    minHeight: 42,
    marginHorizontal: 3,
    borderRadius: 8,
  },
  activeActionButton: {
    backgroundColor: '#e7f0ff',
  },
  pressedActionButton: {
    opacity: 0.62,
  },
  actionText: {
    color: '#4b5568',
    fontSize: 13,
    fontWeight: '700',
  },
  actionIcon: {
    width: 18,
    height: 18,
    marginRight: 5,
    tintColor: '#4b5568',
    resizeMode: 'contain',
  },
  activeActionIcon: {
    tintColor: '#1264d7',
  },
  activeActionText: {
    color: '#1264d7',
  },
});
```

Màn hình kết quả





Social Media Feed

Connect, share, and enjoy



392 Likes 67 Comments 31 Shares

Likes Comments Shares

Bao Tran
Just now

Weekend hike completed. Tired legs, clear mind, and a wonderful view.



176 Likes 29 Comments 12 Shares

Liked Comments Shares

Social Media Feed

Connect, share, and enjoy

Linh Nguyen
Just now

A quiet morning by the lake is the best way to reset for a new week.



129 Likes 28 Comments 13 Shares

Liked Comments Shares

Minh Anh
Just now

Found a colorful corner of the city today. Every street has a story.

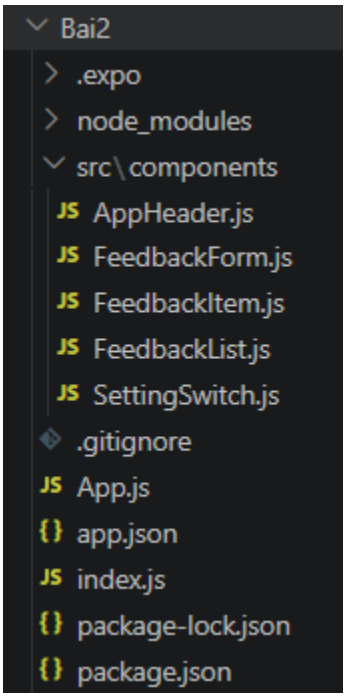


Bài 2: Settings and Feedback

Link video

https://drive.google.com/file/d/1ytCcN_cI2DxY-31Q0TWXy66bxeOdIm79/view?usp=drive_link

Cấu trúc project



Các file code

App.js

Bai2 > JS App.js > ...

```
// 23521720 - Nguyen Minh Tuan

import { useState } from 'react';
import { Alert, ScrollView, StyleSheet, View } from 'react-native';
import { SafeAreaProvider, SafeAreaView } from 'react-native-safe-area-context';
import { StatusBar as ExpoStatusBar } from 'expo-status-bar';

import AppHeader from './src/components/AppHeader';
import FeedbackForm from './src/components/FeedbackForm';
import FeedbackList from './src/components/FeedbackList';
import SettingSwitch from './src/components/SettingSwitch';

export default function App() {
  const [isDarkMode, setIsDarkMode] = useState(false);
  const [notificationsEnabled, setNotificationsEnabled] = useState(true);
  const [feedbackText, setFeedbackText] = useState('');
  const [feedbacks, setFeedbacks] = useState([]);

  const handleSubmit = () => {
    const trimmedFeedback = feedbackText.trim();
    if (!trimmedFeedback) return;

    const newFeedback = {
      id: `${Date.now()}-${feedbacks.length}`,
      content: trimmedFeedback,
    };

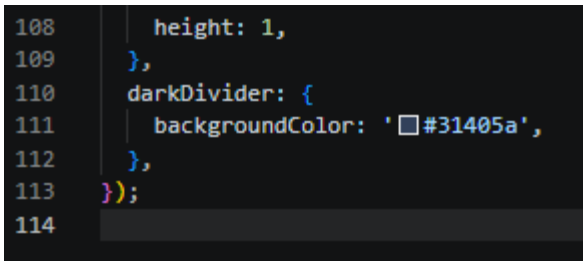
    setFeedbacks((currentFeedbacks) => [newFeedback, ...currentFeedbacks]);
    setFeedbackText('');

    if (notificationsEnabled) {
      Alert.alert('Thank you!', 'Your feedback has been submitted.');
    }
  };

  return (
    <SafeAreaProvider>
      <SafeAreaView style={[styles.safeArea, isDarkMode && styles.darkBackground]}>
        <ExpoStatusBar style={isDarkMode ? 'light' : 'dark'} />
        <ScrollView
          contentContainerStyle={styles.scrollContent}
          keyboardShouldPersistTaps="handled"
          showsVerticalScrollIndicator={false}
        >
          <AppHeader isDarkMode={isDarkMode} />
          <View style={[styles.card, isDarkMode && styles.darkCard]}>
            <SettingSwitch
              label="Dark Mode"
              description="Use a comfortable dark colour theme"
              value={isDarkMode}
              onValueChange={setIsDarkMode}
              isDarkMode={isDarkMode}
            />
            <View style={[styles.divider, isDarkMode && styles.darkDivider]} />
```

```
            <SettingSwitch
              label="Notifications"
              description="Show a confirmation after sending feedback"
              value={notificationsEnabled}
              onValueChange={setNotificationsEnabled}
              isDarkMode={isDarkMode}
            />
          </View>
          <FeedbackForm
            value={feedbackText}
            onChangeText={setFeedbackText}
            onSubmit={handleSubmit}
            isDarkMode={isDarkMode}
          />
          <FeedbackList feedbacks={feedbacks} isDarkMode={isDarkMode} />
        </ScrollView>
      </SafeAreaView>
    </SafeAreaProvider>
  );
}

const styles = StyleSheet.create({
  safeArea: {
    flex: 1,
    backgroundColor: '#f3f6fb',
  },
  darkBackground: {
    backgroundColor: '#0d1321',
  },
  scrollContent: {
    flexGrow: 1,
    paddingBottom: 36,
    paddingHorizontal: 18,
  },
  card: {
    backgroundColor: '#ffffff',
    borderColor: '#e1e7f0',
    borderRadius: 18,
    borderWidth: 1,
    elevation: 2,
    marginBottom: 22,
    paddingHorizontal: 16,
    shadowColor: '#10233f',
    shadowOffset: { width: 0, height: 4 },
    shadowOpacity: 0.06,
    shadowRadius: 10,
  },
  darkCard: {
    backgroundColor: '#182235',
    borderColor: '#2b3b55',
  },
  divider: {
    backgroundColor: '#e8edf4',
```



```
    height: 1,
  },
  darkDivider: {
    backgroundColor: '#31405a',
  },
});

```

AppHeader.js

Bai2 > src > components > JS AppHeader.js > ...

```
// 23521720 - Nguyen Minh Tuan

import { StyleSheet, Text, View } from 'react-native';

export default function AppHeader({ isDarkMode }) {
  return (
    <View style={styles.container}>
      <View style={[styles.logo, isDarkMode && styles.darkLogo]}>
        <Text style={styles.logoMark}>⚛</Text>
      </View>
      <Text style={[styles.title, isDarkMode && styles.lightText]}>React Native App</Text>
      <Text style={[styles.subtitle, isDarkMode && styles.darkSubtitle]}>
        Settings & Feedback
      </Text>
    </View>
  );
}

const styles = StyleSheet.create({
  container: {
    alignItems: 'center',
    paddingBottom: 24,
    paddingTop: 24,
  },
  logo: {
    alignItems: 'center',
    backgroundColor: '#10233f',
    borderRadius: 42,
    height: 84,
    justifyContent: 'center',
    marginBottom: 12,
    width: 84,
  },
  darkLogo: {
    backgroundColor: '#22304a',
  },
  logoMark: {
    color: '#61dafb',
    fontSize: 50,
  },
  title: {
    color: '#14213d',
    fontSize: 24,
    fontWeight: '800',
  },
  lightText: {
    color: '#f8fafc',
  },
  subtitle: {
    color: '#66748a',
    fontSize: 14,
    marginTop: 4,
  },
  darkSubtitle: {
```

```
    color: '#a9b7cc',
  },
});

```

SettingSwitch.js

Bai2 > src > components > JS SettingSwitch.js > ...

```
// 23521720 - Nguyen Minh Tuan
import { StyleSheet, Switch, Text, View } from 'react-native';

export default function SettingSwitch({ label, description, value, onValueChange, isDarkMode }) {
  return (
    <View style={styles.container}>
      <View style={styles.copy}>
        <Text style={[styles.label, isDarkMode && styles.lightText]}>{label}</Text>
        <Text style={[styles.description, isDarkMode && styles.darkDescription]}>{description}</Text>
      </View>
      <Switch
        accessibilityLabel={label}
        value={value}
        onValueChange={onValueChange}
        trackColor={switchTrackColors}
        thumbColor={value ? '#ffffff' : '#f1f5f9'}
      />
    </View>
  );
}

const switchTrackColors = {
  false: '#b8c2d1',
  true: '#14b8a6',
};

const styles = StyleSheet.create({
  container: {
    alignItems: 'center',
    flexDirection: 'row',
    minHeight: 80,
  },
  copy: {
    flex: 1,
    paddingRight: 14,
  },
  label: {
    color: '#17243d',
    fontSize: 16,
    fontWeight: '700',
  },
  lightText: {
    color: '#f8fafc',
  },
  description: {
    color: '#718096',
    fontSize: 12,
    lineHeight: 17,
    marginTop: 3,
  },
  darkDescription: {
    color: '#a9b7cc',
  },
});
```

FeedbackForm.js

Bai2 > src > components > JS FeedbackForm.js > ...

```
// 23521720 - Nguyen Minh Tuan

import { Pressable, StyleSheet, Text, TextInput, View } from 'react-native';

export default function FeedbackForm({ value, onChangeText, onSubmit, isDarkMode }) {
  return (
    <View style={styles.container}>
      <Text style={[styles.title, isDarkMode && styles.lightText]}>Feedback</Text>
      <Text style={[styles.helper, isDarkMode && styles.darkHelper]}>
        Tell us what would make your experience better.
      </Text>
      <TextInput
        accessibilityLabel="Feedback"
        multiline
        numberOfLines={5}
        onChangeText={onChangeText}
        placeholder="Your feedback here..."
        placeholderTextColor={isDarkMode ? '#8fa0b8' : '#8995a8'}
        style={[styles.input, isDarkMode && styles.darkInput]}
        textAlignVertical="top"
        value={value}
      />
      <Pressable
        accessibilityRole="button"
        onPress={onSubmit}
        style={({ pressed }) => [styles.button, pressed && styles.buttonPressed]}
      >
        <Text style={styles.buttonText}>Send Feedback</Text>
      </Pressable>
    </View>
  );
}

const styles = StyleSheet.create({
  container: {
    marginBottom: 26,
  },
  title: {
    color: '#14213d',
    fontSize: 24,
    fontWeight: '800',
  },
  lightText: {
    color: '#f8fafc',
  },
  helper: {
    color: '#66748a',
    fontSize: 13,
    marginBottom: 12,
    marginTop: 4,
  },
  darkHelper: {
    color: '#a9b7cc',
  },
```

```
  input: {
    backgroundColor: '#ffffff',
    borderColor: '#cfd8e6',
    borderRadius: 14,
    borderWidth: 1,
    color: '#17243d',
    fontSize: 15,
    minHeight: 130,
    paddingHorizontal: 14,
    paddingVertical: 13,
  },
  darkInput: {
    backgroundColor: '#182235',
    borderColor: '#3a4a64',
    color: '#f8fafc',
  },
  button: {
    alignItems: 'center',
    backgroundColor: '#2563eb',
    borderRadius: 12,
    elevation: 2,
    justifyContent: 'center',
    marginTop: 12,
    minHeight: 50,
    shadowColor: '#1d4ed8',
    shadowOffset: { width: 0, height: 3 },
    shadowOpacity: 0.22,
    shadowRadius: 7,
  },
  buttonPressed: {
    backgroundColor: '#1d4ed8',
  },
  buttonText: {
    color: '#ffffff',
    fontSize: 15,
    fontWeight: '800',
    letterSpacing: 0.4,
  },
});

```

FeedbackList.js

Bai2 > src > components > JS FeedbackList.js > ...

```
// 23521720 - Nguyen Minh Tuan

import { StyleSheet, Text, View } from 'react-native';

import FeedbackItem from './FeedbackItem';

export default function FeedbackList({ feedbacks, isDarkMode }) {
  return (
    <View style={styles.container}>
      <Text style={[styles.title, isDarkMode && styles.lightText]}>Frequently Asked Questions</Text>
      {feedbacks.length === 0 ? (
        <View style={[styles.emptyState, isDarkMode && styles.darkEmptyState]}>
          <Text style={styles.emptyIcon}>✦</Text>
          <Text style={[styles.emptyText, isDarkMode && styles.darkEmptyText]}>
            Submitted feedback will appear here.
          </Text>
        </View>
      ) : (
        feedbacks.map((feedback) => (
          <FeedbackItem key={feedback.id} feedback={feedback} isDarkMode={isDarkMode} />
        ))
      )}
    </View>
  );
}

const styles = StyleSheet.create({
  container: {
    paddingBottom: 20,
  },
  title: {
    color: '#14213d',
    fontSize: 22,
    fontWeight: '800',
    marginBottom: 12,
  },
  lightText: {
    color: '#f8fafc',
  },
  emptyState: {
    alignItems: 'center',
    backgroundColor: '#eaf1fb',
    borderRadius: 14,
    paddingHorizontal: 20,
    paddingVertical: 24,
  },
  darkEmptyState: {
    backgroundColor: '#182235',
  },
  emptyIcon: {
    color: '#2563eb',
    fontSize: 24,
    marginBottom: 6,
  },
```



```
  emptyText: {
    color: '#66748a',
    fontSize: 14,
    textAlign: 'center',
  },
  darkEmptyText: {
    color: '#a9b7cc',
  },
});

```

FeedbackItem.js

Bai2 > src > components > JS FeedbackItem.js > ...

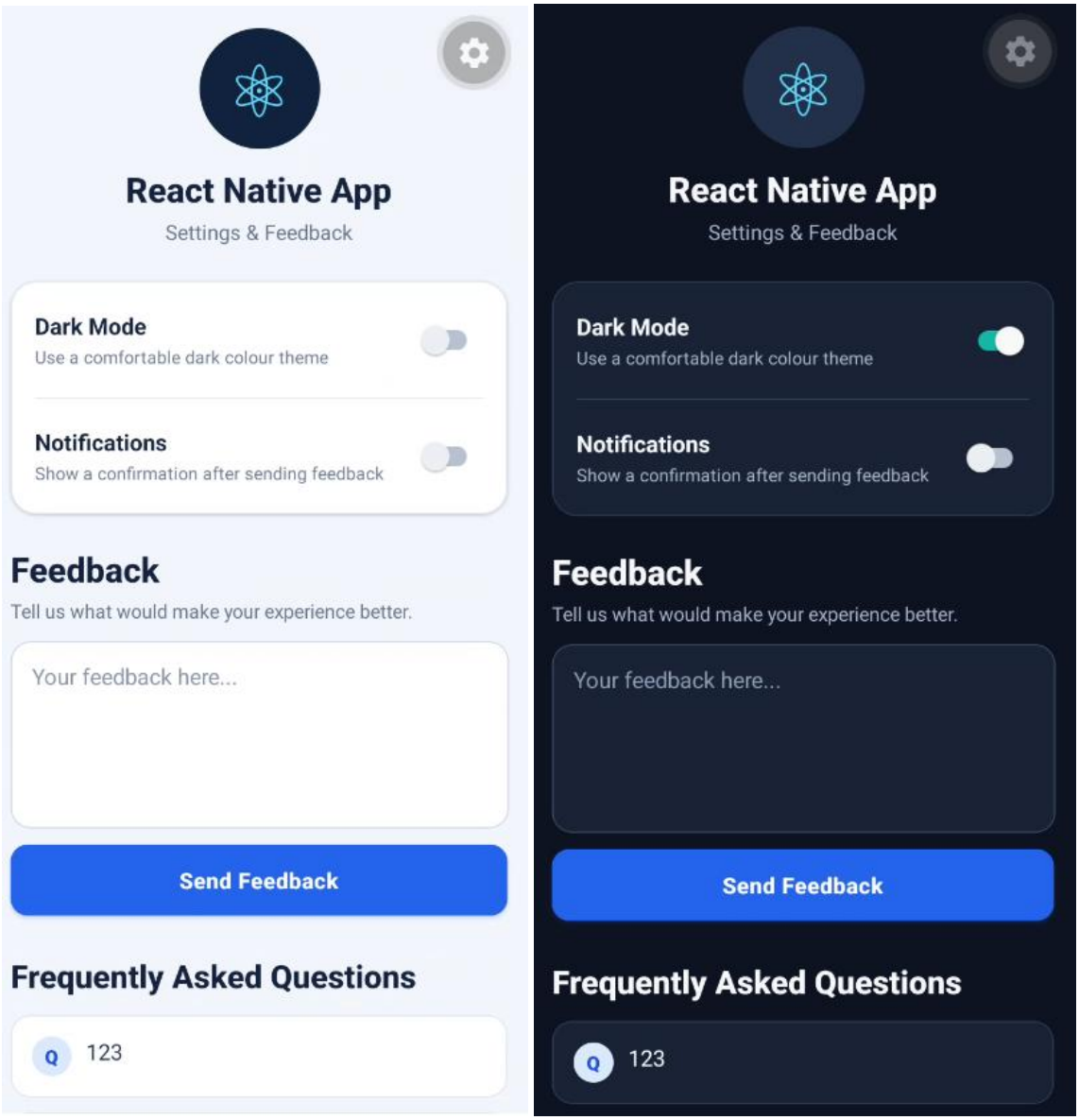
```
// 23521720 - Nguyen Minh Tuan

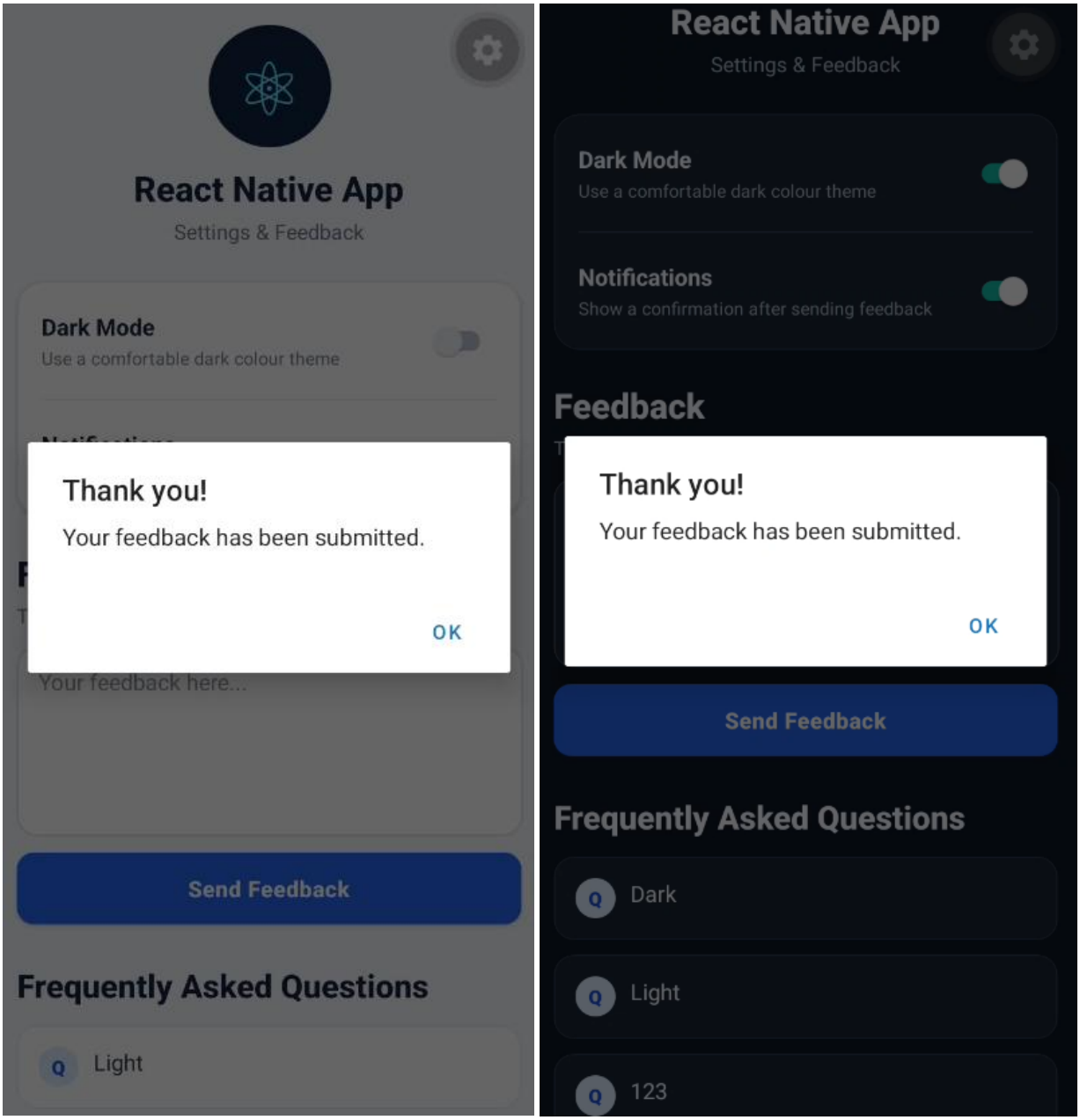
import { StyleSheet, Text, View } from 'react-native';

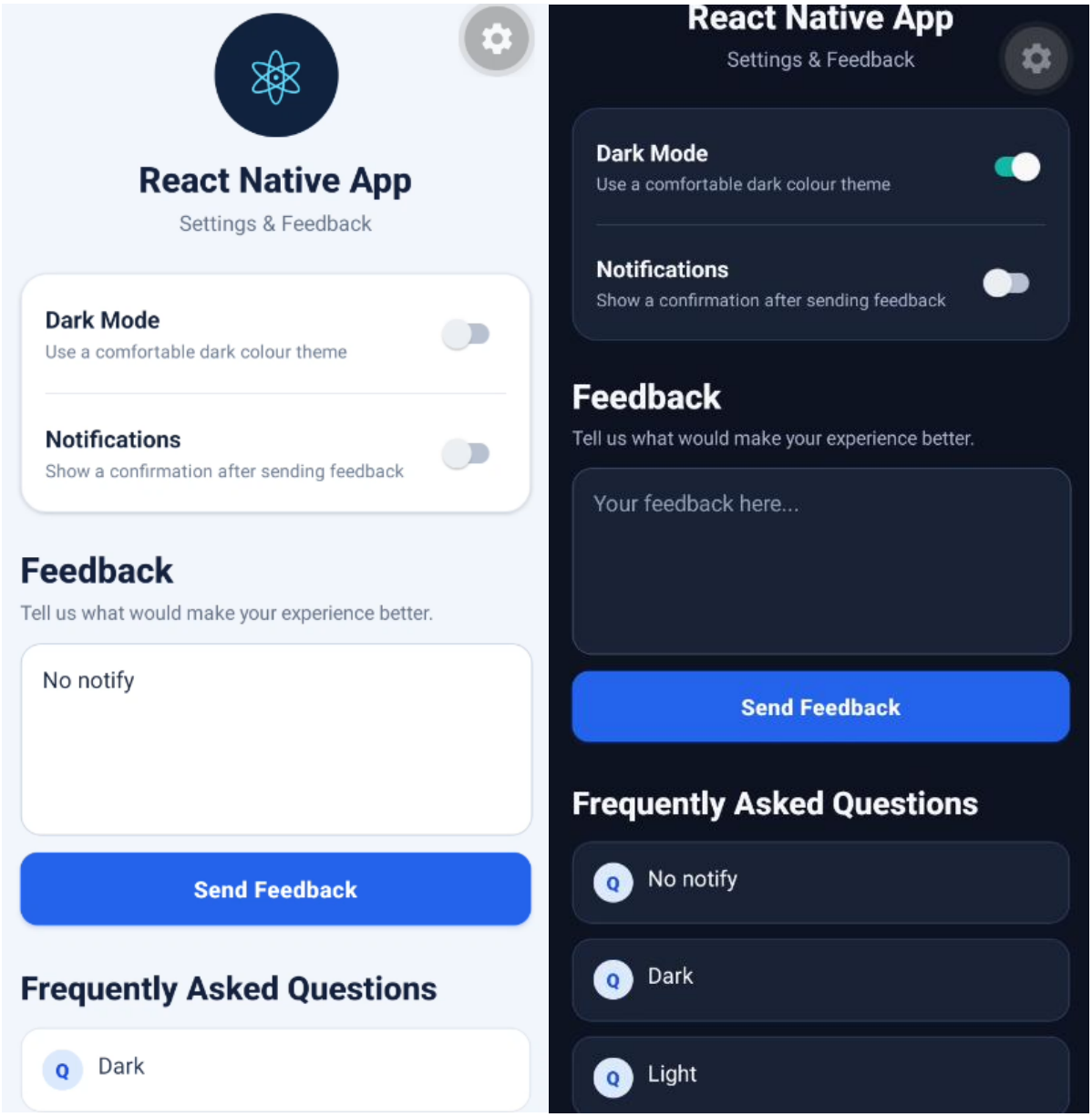
export default function FeedbackItem({ feedback, isDarkMode }) {
  return (
    <View style={[styles.card, isDarkMode && styles.darkCard]}>
      <View style={styles.badge}>
        <Text style={styles.badgeText}>Q</Text>
      </View>
      <Text style={[styles.content, isDarkMode && styles.lightText]}>{feedback.content}</Text>
    </View>
  );
}

const styles = StyleSheet.create({
  card: {
    alignItems: 'flex-start',
    backgroundColor: '#ffffff',
    borderColor: '#e1e7f0',
    borderRadius: 14,
    borderWidth: 1,
    flexDirection: 'row',
    marginBottom: 10,
    padding: 14,
  },
  darkCard: {
    backgroundColor: '#182235',
    borderColor: '#2b3b55',
  },
  badge: {
    alignItems: 'center',
    backgroundColor: '#dbeafe',
    borderRadius: 14,
    height: 28,
    justifyContent: 'center',
    marginRight: 10,
    width: 28,
  },
  badgeText: {
    color: '#1d4ed8',
    fontSize: 13,
    fontWeight: '900',
  },
  content: {
    color: '#263651',
    flex: 1,
    fontSize: 15,
    lineHeight: 22,
  },
  lightText: {
    color: '#f8fafc',
  },
});
```

Màn hình kết quả




React Native App
Settings & Feedback
Dark Mode
Use a comfortable dark colour theme
Notifications
Show a confirmation after sending feedback
Feedback
Thank you!
Your feedback has been submitted.
OK
Send Feedback
Frequently Asked Questions
Q Dark
Q Light
Q 123

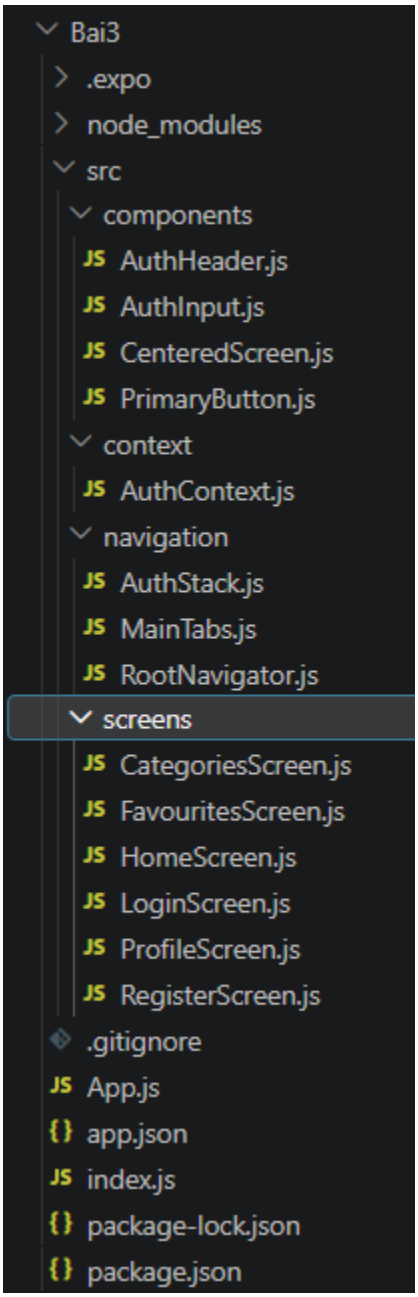

React Native App
Settings & Feedback
Dark Mode
Use a comfortable dark colour theme
Notifications
Show a confirmation after sending feedback
Feedback
Tell us what would make your experience better.
No notify
Send Feedback
Frequently Asked Questions
Q
Dark
React Native App
Settings & Feedback
Dark Mode
Use a comfortable dark colour theme
Notifications
Show a confirmation after sending feedback
Feedback
Tell us what would make your experience better.
Your feedback here...
Send Feedback
Frequently Asked Questions
Q
No notify
Q
Dark
Q
Light

Bài 3: React Navigation

Link video

https://drive.google.com/file/d/1nXdKMDCb4BnohbIx863OOG5KJYDrEwLu/view?usp=drive_link

Cấu trúc project



Các file code

App.js

Bai3 > JS App.js > ...

```
1  // 23521720 - Nguyen Minh Tuan
2
3  import { SafeAreaProvider } from 'react-native-safe-area-context';
4  import { StatusBar } from 'expo-status-bar';
5
6  import { AuthProvider } from './src/context/AuthContext';
7  import RootNavigator from './src/navigation/RootNavigator';
8
9  export default function App() {
10   return (
11     <SafeAreaProvider>
12       <AuthProvider>
13         <StatusBar style="dark" />
14         <RootNavigator />
15       </AuthProvider>
16     </SafeAreaProvider>
17   );
18 }
19
```

AuthContext.js

Bai3 > src > context > JS AuthContext.js > ...

```
// 23521720 - Nguyen Minh Tuan

import { createContext, useContext, useEffect, useMemo, useState } from 'react';

const VALID_EMAIL = '23521720@gm.uit.edu.vn';
const VALID_PASSWORD = 'nguyenminhtuan';

const AuthContext = createContext(undefined);

export function AuthProvider({ children }) {
  const [isAuthenticated, setIsAuthenticated] = useState(false);
  const [isChecking, setIsChecking] = useState(true);

  useEffect(() => {
    setIsChecking(false);
  }, []);

  const value = useMemo(
    () => ({
      isAuthenticated,
      isChecking,
      login: (email, password) => {
        const isValid = email.trim().toLowerCase() === VALID_EMAIL && password === VALID_PASSWORD;
        if (isValid) setIsAuthenticated(true);
        return isValid;
      },
      logout: () => setIsAuthenticated(false),
    }),
    [isAuthenticated, isChecking],
  );

  return <AuthContext.Provider value={value}>{children}</AuthContext.Provider>;
}

export function useAuth() {
  const context = useContext(AuthContext);
  if (!context) throw new Error('useAuth must be used inside AuthProvider.');
  return context;
}

```

RootNavigator.js

Bai3 > src > navigation > JS RootNavigator.js > ...

```
// 23521720 - Nguyen Minh Tuan

import { ActivityIndicator, StyleSheet, Text, View } from 'react-native';
import { NavigationContainer } from '@react-navigation/native';

import { useAuth } from '../context/AuthContext';
import AuthStack from './AuthStack';
import MainTabs from './MainTabs';

export default function RootNavigator() {
  const { isAuthenticated, isChecking } = useAuth();

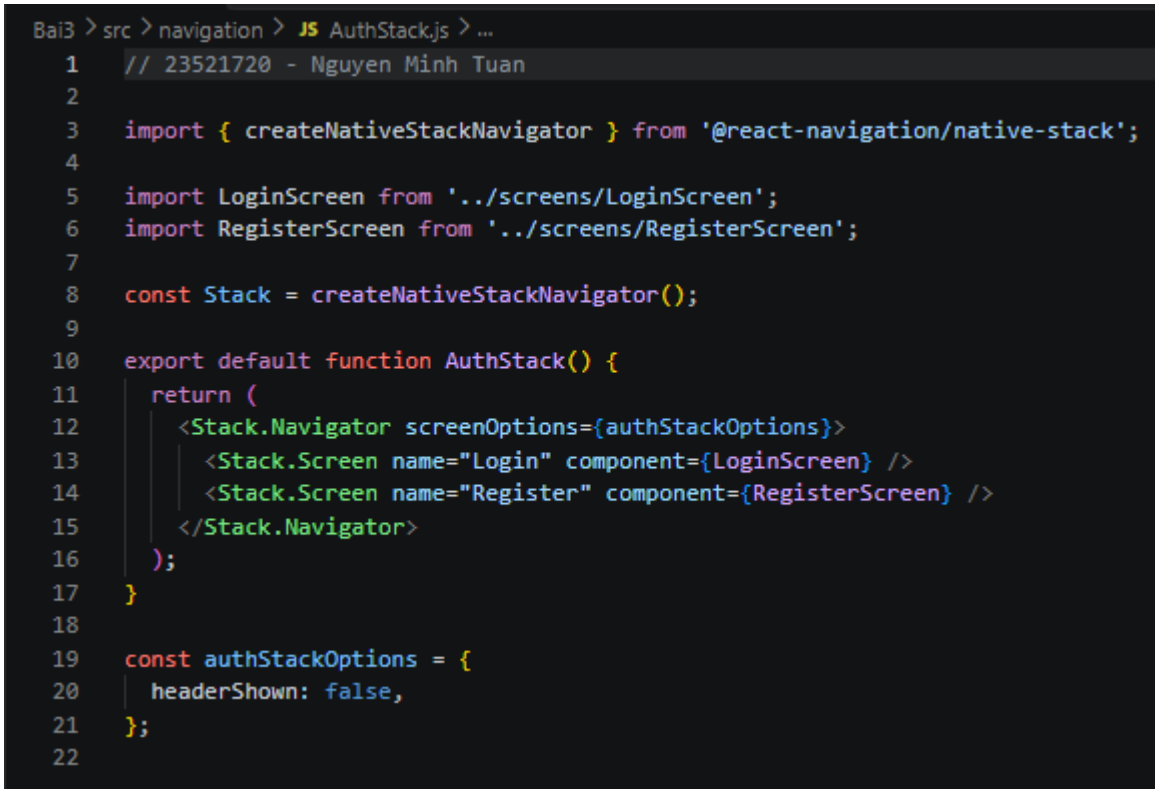
  if (isChecking) {
    return (
      <View style={styles.loadingContainer}>
        <ActivityIndicator color="#ea580c" size="large" />
        <Text style={styles.loadingText}>Checking session...</Text>
      </View>
    );
  }

  return (
    <NavigationContainer>
      {isAuthenticated ? <MainTabs /> : <AuthStack />}
    </NavigationContainer>
  );
}

const styles = StyleSheet.create({
  loadingContainer: {
    alignItems: 'center',
    backgroundColor: '#f8fafc',
    flex: 1,
    justifyContent: 'center',
  },
  loadingText: {
    color: '#68768a',
    fontSize: 14,
    marginTop: 12,
  },
});

```

AuthStack.js



MainTabs.js

Bai3 > src > navigation > JS MainTabs.js > ...

```
// 23521720 - Nguyen Minh Tuan

import { useMemo } from 'react';
import { StyleSheet, Text } from 'react-native';
import { createBottomTabNavigator } from '@react-navigation/bottom-tabs';
import { useSafeAreaInsets } from 'react-native-safe-area-context';

import CategoriesScreen from '../screens/CategoriesScreen';
import FavouritesScreen from '../screens/FavouritesScreen';
import HomeScreen from '../screens/HomeScreen';
import ProfileScreen from '../screens/ProfileScreen';

const Tab = createBottomTabNavigator();

const tabIcons = {
  Home: '⌂',
  Categories: '▦',
  Favourites: '♥',
  Profile: '●',
};

export default function MainTabs() {
  const { bottom } = useSafeAreaInsets();
  const tabBarStyles = useMemo(() => createTabBarStyles(bottom), [bottom]);

  return (
    <Tab.Navigator
      screenOptions={({ route }) => ({
        headerShown: false,
        tabBarActiveTintColor: '■#2563eb',
        tabBarInactiveTintColor: '■#718096',
        tabBarBadge: route.name === 'Favourites' ? 3 : undefined,
        tabBarBadgeStyle: styles.badge,
        tabBarLabelStyle: styles.label,
        tabBarLabelPosition: 'below-icon',
        tabBarStyle: tabBarStyles.tabBar,
        tabBarIcon: ({ focused }) => (
          <Text style={[styles.icon, focused ? styles.activeIcon : styles.inactiveIcon]}>{tabIcons[route.name]}</Text>
        ),
      })}
    >
      <Tab.Screen name="Home" component={HomeScreen} />
      <Tab.Screen name="Categories" component={CategoriesScreen} />
      <Tab.Screen name="Favourites" component={FavouritesScreen} />
      <Tab.Screen name="Profile" component={ProfileScreen} />
    </Tab.Navigator>
  );
}

function createTabBarStyles(bottomInset) {
  return StyleSheet.create({
    tabBar: {
      backgroundColor: '■#ffffff',
      borderTopColor: '■#e2e8f0',
```

```
        // Reserve the system navigation area and keep room for both icon and label
        height: 64 + bottomInset,
        paddingTop: 7,
        paddingBottom: bottomInset + 8,
      },
    });
  }

  const styles = StyleSheet.create({
    label: {
      fontSize: 11,
      fontWeight: '700',
    },
    icon: {
      fontSize: 24,
      fontWeight: '800',
    },
    badge: {
      backgroundColor: '#ef4444',
      color: '#ffffff',
      fontSize: 11,
    },
    activeIcon: {
      color: '#2563eb',
    },
    inactiveIcon: {
      color: '#718096',
    },
  });

```

LoginScreen.js

Bai3 > src > screens > JS LoginScreen.js > ...

```
// 23521720 - Nguyen Minh Tuan

import { useState } from 'react';
import {
  Alert,
  KeyboardAvoidingView,
  Platform,
  Pressable,
  ScrollView,
  StyleSheet,
  Text,
  View,
} from 'react-native';
import { SafeAreaView } from 'react-native-safe-area-context';

import AuthHeader from '../components/AuthHeader';
import AuthInput from '../components/AuthInput';
import PrimaryButton from '../components/PrimaryButton';
import { useAuth } from '../context/AuthContext';

export default function LoginScreen({ navigation }) {
  const [email, setEmail] = useState('');
  const [password, setPassword] = useState('');
  const { login } = useAuth();

  const handleLogin = () => {
    if (!login(email, password)) {
      Alert.alert('Login failed', 'Incorrect email or password.');
    }
  };

  return (
    <SafeAreaView style={styles.safeArea}>
      <KeyboardAvoidingView
        behavior={Platform.OS === 'ios' ? 'padding' : 'height'}
        style={styles.keyboardView}
      >
        <ScrollView
          contentContainerStyle={styles.scrollContent}
          keyboardShouldPersistTaps="handled"
          showsVerticalScrollIndicator={false}
        >
          <View style={styles.form}>
            <AuthHeader title="Welcome" subtitle="Sign in to continue to your app" />
            <AuthInput
              autoCapitalize="none"
              autoCorrect={false}
              icon="✉"
              keyboardType="email-address"
              onChangeText={setEmail}
              placeholder="Email"
              value={email}
            />
            <AuthInput
```

```
55                icon="▫"
56                onChangeText={setPassword}
57                placeholder="Password"
58                secureTextEntry
59                value={password}
60              />
61              <PrimaryButton title="LOG IN" onPress={handleLogin} />
62              <View style={styles.linkRow}>
63                <Text style={styles.linkPrefix}>New here? </Text>
64                <Pressable onPress={() => navigation.navigate('Register')}>
65                  <Text style={styles.link}>Sign up here!</Text>
66                </Pressable>
67              </View>
68            </View>
69          </ScrollView>
70        </KeyboardAvoidingView>
71      </SafeAreaView>
72    );
73  }
74
75  const styles = StyleSheet.create({
76    safeArea: {
77      backgroundColor: '#f7f9fc',
78      flex: 1,
79    },
80    keyboardView: {
81      flex: 1,
82    },
83    scrollContent: {
84      flexGrow: 1,
85      justifyContent: 'center',
86      paddingHorizontal: 24,
87      paddingVertical: 30,
88    },
89    form: {
90      alignSelf: 'center',
91      maxWidth: 520,
92      width: '100%',
93    },
94    linkRow: {
95      alignItems: 'center',
96      flexDirection: 'row',
97      flexWrap: 'wrap',
98      justifyContent: 'center',
99      marginTop: 24,
100   },
101   linkPrefix: {
102     color: '#64748b',
103     fontSize: 14,
104   },
105   link: {
106     color: '#2563eb',
107     fontSize: 14,
```

```
      fontWeight: '800',
    },
  });

```

RegisterScreen.js

Bai3 > src > screens > JS RegisterScreen.js > ...

```
// 23521720 - Nguyen Minh Tuan

import { useState } from 'react';
import {
  Alert,
  KeyboardAvoidingView,
  Platform,
  Pressable,
  ScrollView,
  StyleSheet,
  Text,
  View,
} from 'react-native';
import { SafeAreaView } from 'react-native-safe-area-context';

import AuthHeader from '../components/AuthHeader';
import AuthInput from '../components/AuthInput';
import PrimaryButton from '../components/PrimaryButton';

export default function RegisterScreen({ navigation }) {
  const [username, setUsername] = useState('');
  const [email, setEmail] = useState('');
  const [password, setPassword] = useState('');

  const handleCreate = () => {
    Alert.alert('Registration demo', 'Account creation is not implemented in this lab.');
  };

  return (
    <SafeAreaView style={styles.safeArea}>
      <KeyboardAvoidingView
        behavior={Platform.OS === 'ios' ? 'padding' : 'height'}
        style={styles.keyboardView}
      >
        <ScrollView
          contentContainerStyle={styles.scrollContent}
          keyboardShouldPersistTaps="handled"
          showsVerticalScrollIndicator={false}
        >
          <View style={styles.form}>
            <AuthHeader title="Create New Account" subtitle="Registration UI for navigation practice" />
            <AuthInput
              autoCapitalize="words"
              icon="ℹ"
              onChangeText={setUsername}
              placeholder="Enter username"
              value={username}
            />
            <AuthInput
              autoCapitalize="none"
              autoCorrect={false}
              icon="✉"
              keyboardType="email-address"
              onChangeText={setEmail}
```

```
            placeholder="Enter email"
            value={email}
          />
          <AuthInput
            icon="▫"
            onChangeText={setPassword}
            placeholder="Enter password"
            secureTextEntry
            value={password}
          />
          <PrimaryButton title="CREATE" onPress={handleCreate} />
          <View style={styles.linkRow}>
            <Text style={styles.linkPrefix}>Already have an account? </Text>
            <Pressable onPress={() => navigation.goBack()}>
              <Text style={styles.link}>Login now!</Text>
            </Pressable>
          </View>
        </View>
      </ScrollView>
    </KeyboardAvoidingView>
  </SafeAreaView>
 );
}

const styles = StyleSheet.create({
  safeArea: {
    backgroundColor: '#f7f9fc',
    flex: 1,
  },
  keyboardView: {
    flex: 1,
  },
  scrollContent: {
    flexGrow: 1,
    justifyContent: 'center',
    paddingHorizontal: 24,
    paddingVertical: 30,
  },
  form: {
    alignSelf: 'center',
    maxWidth: 520,
    width: '100%',
  },
  linkRow: {
    alignItems: 'center',
    flexDirection: 'row',
    flexWrap: 'wrap',
    justifyContent: 'center',
    marginTop: 24,
  },
  linkPrefix: {
    color: '#64748b',
    fontSize: 14,
```

```
108     },
109     link: {
110       color: '#2563eb',
111       fontSize: 14,
112       fontWeight: '800',
113     },
114   });
115
```

HomeScreen.js

```
Bai3 > src > screens > JS HomeScreen.js > ...
1   // 23521720 - Nguyen Minh Tuan
2
3   import CenteredScreen from '../components/CenteredScreen';
4
5   export default function HomeScreen() {
6     return <CenteredScreen icon="⌂" title="Home Screen" description="Welcome to the main area of the app." />;
7   }
8
```

CategoriesScreen.js

```
Bai3 > src > screens > JS CategoriesScreen.js > ...
1   // 23521720 - Nguyen Minh Tuan
2
3   import CenteredScreen from '../components/CenteredScreen';
4
5   export default function CategoriesScreen() {
6     return <CenteredScreen icon="▦" title="Categories Screen" description="Browse content by category." />;
7   }
8
```

FavouritesScreen.js

```
Bai3 > src > screens > JS FavouritesScreen.js > ...
1   // 23521720 - Nguyen Minh Tuan
2
3   import CenteredScreen from '../components/CenteredScreen';
4
5   export default function FavouritesScreen() {
6     return <CenteredScreen icon="♥" title="Favourites Screen" description="You currently have 3 favourite items." />;
7   }
8
```

ProfileScreen.js

Bai3 > src > screens > JS ProfileScreen.js > ...

```
// 23521720 - Nguyen Minh Tuan

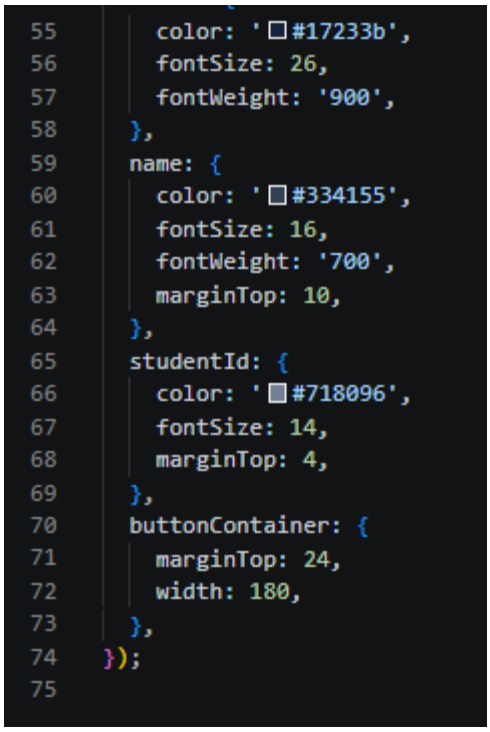
import { StyleSheet, Text, View } from 'react-native';
import { SafeAreaView } from 'react-native-safe-area-context';

import PrimaryButton from '../components/PrimaryButton';
import { useAuth } from '../context/AuthContext';

export default function ProfileScreen() {
  const { logout } = useAuth();

  return (
    <SafeAreaView style={styles.safeArea}>
      <View style={styles.container}>
        <View style={styles.avatar}>
          <Text style={styles.avatarText}>NT</Text>
        </View>
        <Text style={styles.title}>Profile Screen</Text>
        <Text style={styles.name}>Nguyen Minh Tuan</Text>
        <Text style={styles.studentId}>MSSV: 23521720</Text>
        <View style={styles.buttonContainer}>
          <PrimaryButton title="LOG OUT" onPress={logout} />
        </View>
      </View>
    </SafeAreaView>
  );
}

const styles = StyleSheet.create({
  safeArea: {
    backgroundColor: '#f7f9fc',
    flex: 1,
  },
  container: {
    alignItems: 'center',
    flex: 1,
    justifyContent: 'center',
    paddingHorizontal: 28,
  },
  avatar: {
    alignItems: 'center',
    backgroundColor: '#2563eb',
    borderRadius: 44,
    height: 88,
    justifyContent: 'center',
    marginBottom: 18,
    width: 88,
  },
  avatarText: {
    color: '#ffffff',
    fontSize: 26,
    fontWeight: '900',
  },
  title: {
```



```
    color: '#17233b',
    fontSize: 26,
    fontWeight: '900',
  },
  name: {
    color: '#334155',
    fontSize: 16,
    fontWeight: '700',
    marginTop: 10,
  },
  studentId: {
    color: '#718096',
    fontSize: 14,
    marginTop: 4,
  },
  buttonContainer: {
    marginTop: 24,
    width: 180,
  },
});

```

AuthHeader.js

Bai3 > src > components > JS AuthHeader.js > ...

```jsx
// 23521720 - Nguyen Minh Tuan

import { StyleSheet, Text, View } from 'react-native';

export default function AuthHeader({ title, subtitle }) {
  return (
    <View style={styles.container}>
      <View style={styles.logo}>
        <Text style={styles.logoMark}>⚛</Text>
      </View>
      <Text style={styles.title}>{title}</Text>
      <Text style={styles.subtitle}>{subtitle}</Text>
    </View>
  );
}

const styles = StyleSheet.create({
  container: {
    alignItems: 'center',
    marginBottom: 26,
  },
  logo: {
    alignItems: 'center',
    backgroundColor: '#12223d',
    borderRadius: 42,
    height: 84,
    justifyContent: 'center',
    marginBottom: 14,
    width: 84,
  },
  logoMark: {
    color: '#61dafb',
    fontSize: 49,
  },
  title: {
    color: '#17233b',
    fontSize: 30,
    fontWeight: '900',
    textAlign: 'center',
  },
  subtitle: {
    color: '#6b788d',
    fontSize: 14,
    lineHeight: 20,
    marginTop: 6,
    textAlign: 'center',
  },
});

```

AuthInput.js

Bai3 > src > components > JS AuthInput.js > ...

```
// 23521720 - Nguyen Minh Tuan

import { StyleSheet, Text, TextInput, View } from 'react-native';

export default function AuthInput({ icon, ...inputProps }) {
  return (
    <View style={styles.container}>
      <Text style={styles.icon}>{icon}</Text>
      <TextInput
        placeholderTextColor="#8995a8"
        style={styles.input}
        {...inputProps}
      />
    </View>
  );
}

const styles = StyleSheet.create({
  container: {
    alignItems: 'center',
    backgroundColor: '#ffffff',
    borderColor: '#cbd5e1',
    borderRadius: 15,
    borderWidth: 1,
    flexDirection: 'row',
    marginBottom: 14,
    minHeight: 58,
    paddingHorizontal: 16,
  },
  icon: {
    color: '#324158',
    fontSize: 21,
    marginRight: 12,
    textAlign: 'center',
    width: 25,
  },
  input: {
    color: '#17233b',
    flex: 1,
    fontSize: 15,
    minHeight: 56,
  },
});

```

PrimaryButton.js



Bai3 > src > components > JS PrimaryButton.js > ...

```
1   // 23521720 - Nguyen Minh Tuan
2
3   import { Pressable, StyleSheet, Text } from 'react-native';
4
5   export default function PrimaryButton({ title, onPress }) {
6     return (
7       <Pressable
8         accessibilityRole="button"
9         onPress={onPress}
10        style={({ pressed }) => [styles.button, pressed && styles.buttonPressed]}
11      >
12        <Text style={styles.title}>{title}</Text>
13      </Pressable>
14    );
15  }
16
17  const styles = StyleSheet.create({
18    button: {
19      alignItems: 'center',
20      backgroundColor: '#ea580c',
21      borderRadius: 14,
22      elevation: 2,
23      justifyContent: 'center',
24      minHeight: 54,
25      shadowColor: '#9a3412',
26      shadowOffset: { width: 0, height: 4 },
27      shadowOpacity: 0.22,
28      shadowRadius: 8,
29    },
30    buttonPressed: {
31      backgroundColor: '#c2410c',
32    },
33    title: {
34      color: '#ffffff',
35      fontSize: 16,
36      fontWeight: '900',
37      letterSpacing: 0.8,
38    },
39  });
40
```

CenteredScreen.js

Bai3 > src > components > JS CenteredScreen.js > ...

```
// 23521720 - Nguyen Minh Tuan

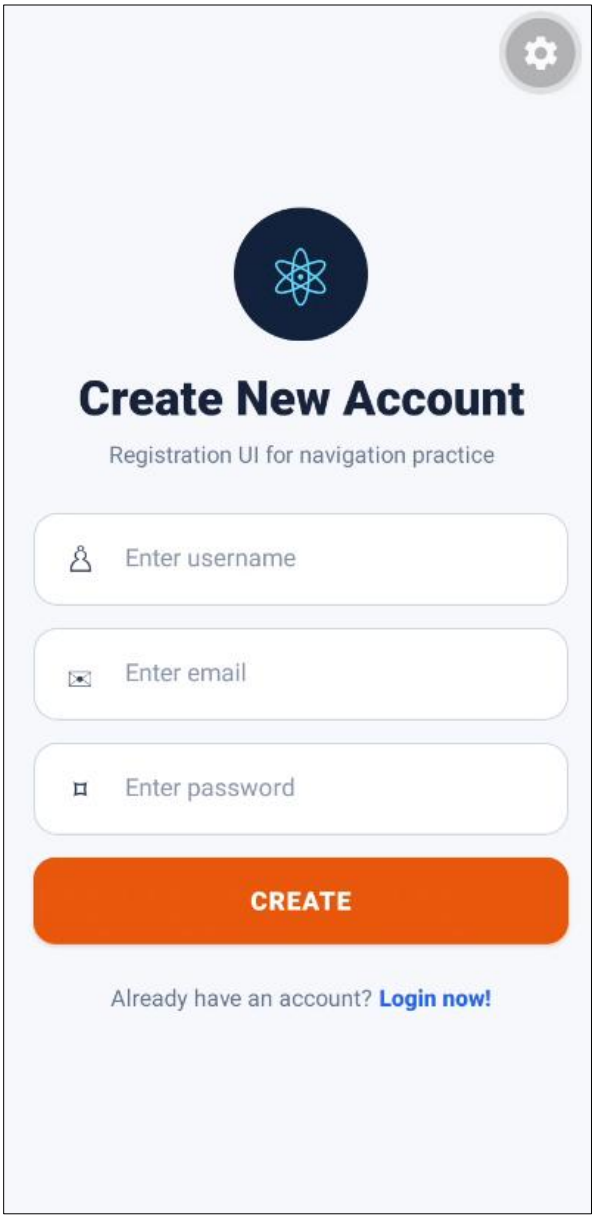
import { StyleSheet, Text, View } from 'react-native';
import { SafeAreaView } from 'react-native-safe-area-context';

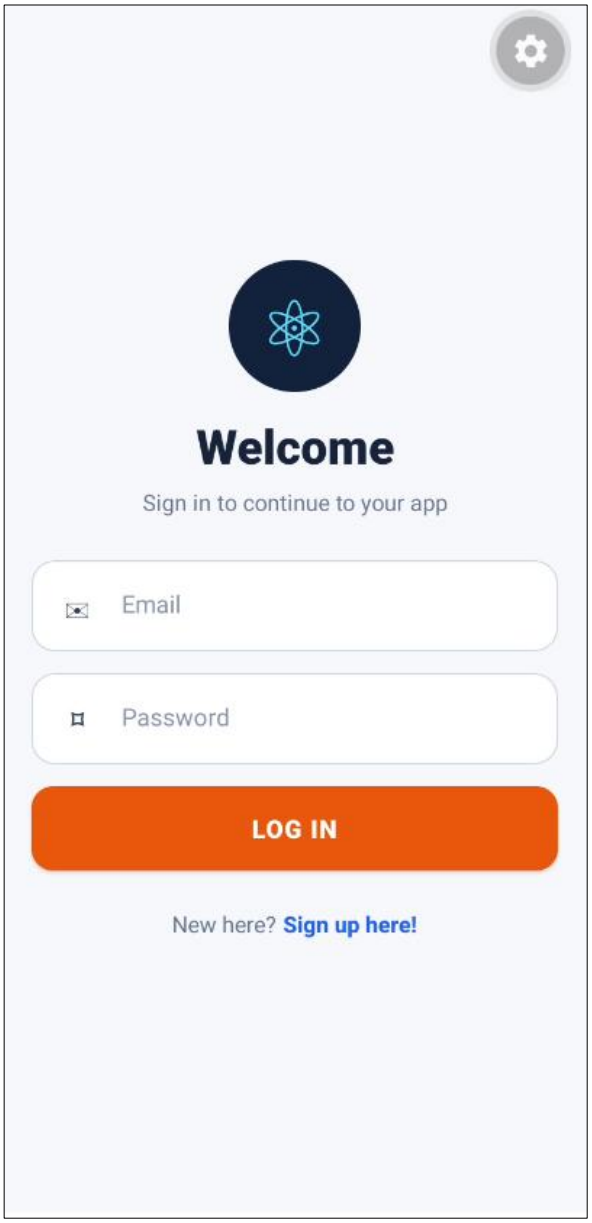
export default function CenteredScreen({ icon, title, description }) {
  return (
    <SafeAreaView style={styles.safeArea}>
      <View style={styles.container}>
        <View style={styles.iconCircle}>
          <Text style={styles.icon}>{icon}</Text>
        </View>
        <Text style={styles.title}>{title}</Text>
        <Text style={styles.description}>{description}</Text>
      </View>
    </SafeAreaView>
  );
}

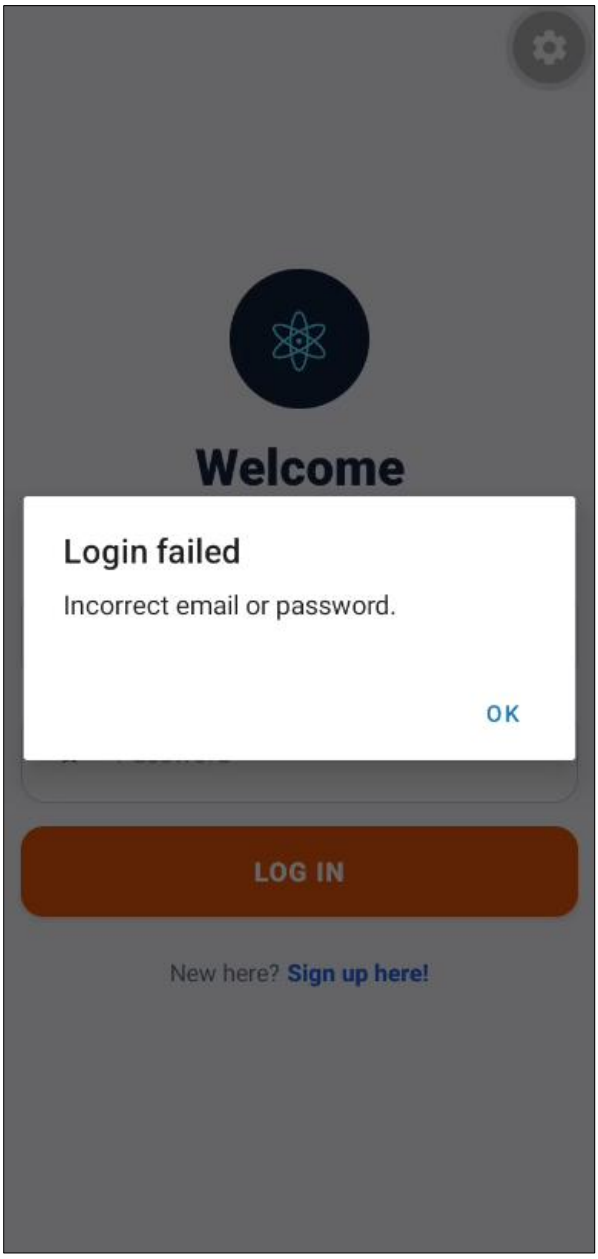
const styles = StyleSheet.create({
  safeArea: {
    backgroundColor: '#f7f9fc',
    flex: 1,
  },
  container: {
    alignItems: 'center',
    flex: 1,
    justifyContent: 'center',
    paddingHorizontal: 28,
  },
  iconCircle: {
    alignItems: 'center',
    backgroundColor: '#dbeafe',
    borderRadius: 38,
    height: 76,
    justifyContent: 'center',
    marginBottom: 18,
    width: 76,
  },
  icon: {
    color: '#2563eb',
    fontSize: 34,
  },
  title: {
    color: '#17233b',
    fontSize: 26,
    fontWeight: '900',
    textAlign: 'center',
  },
  description: {
    color: '#6c7a90',
    fontSize: 14,
    lineHeight: 21,
    marginTop: 8,
```

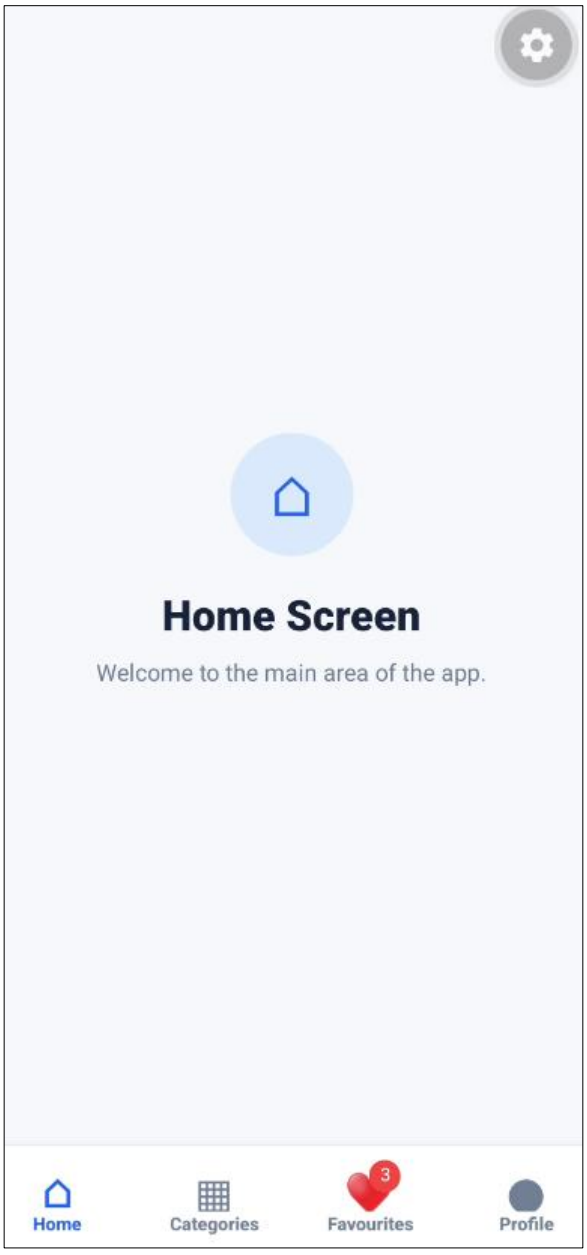
```
55        textAlign: 'center',
56      },
57    });
58
```

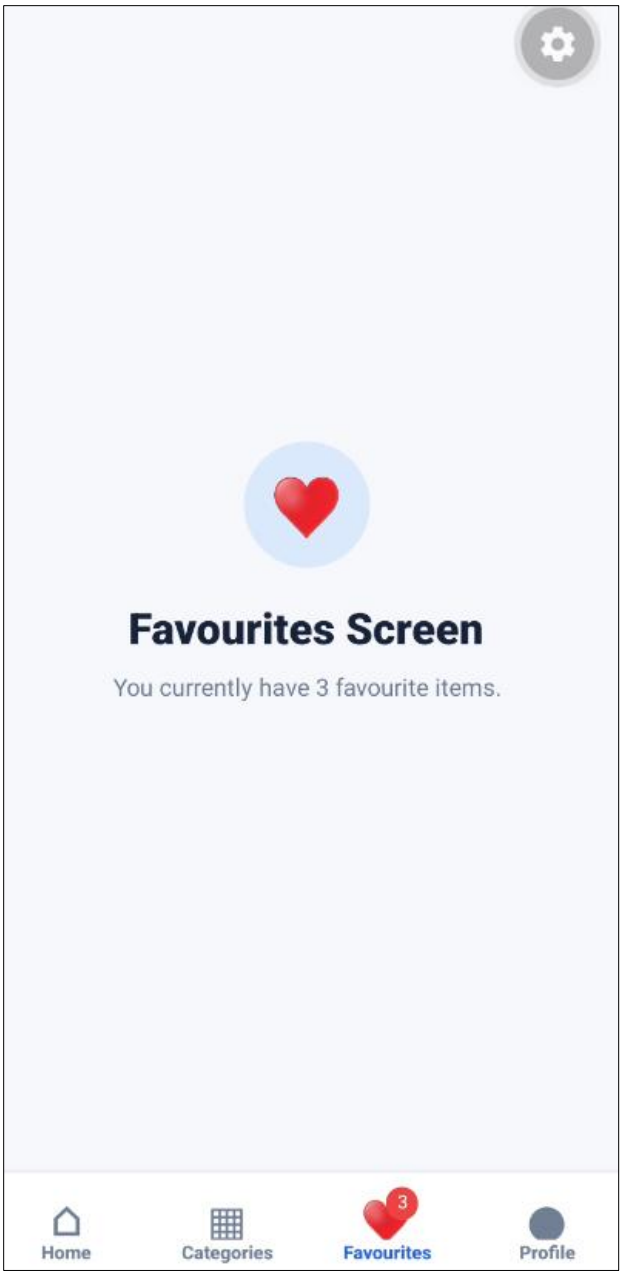
Màn hình kết quả

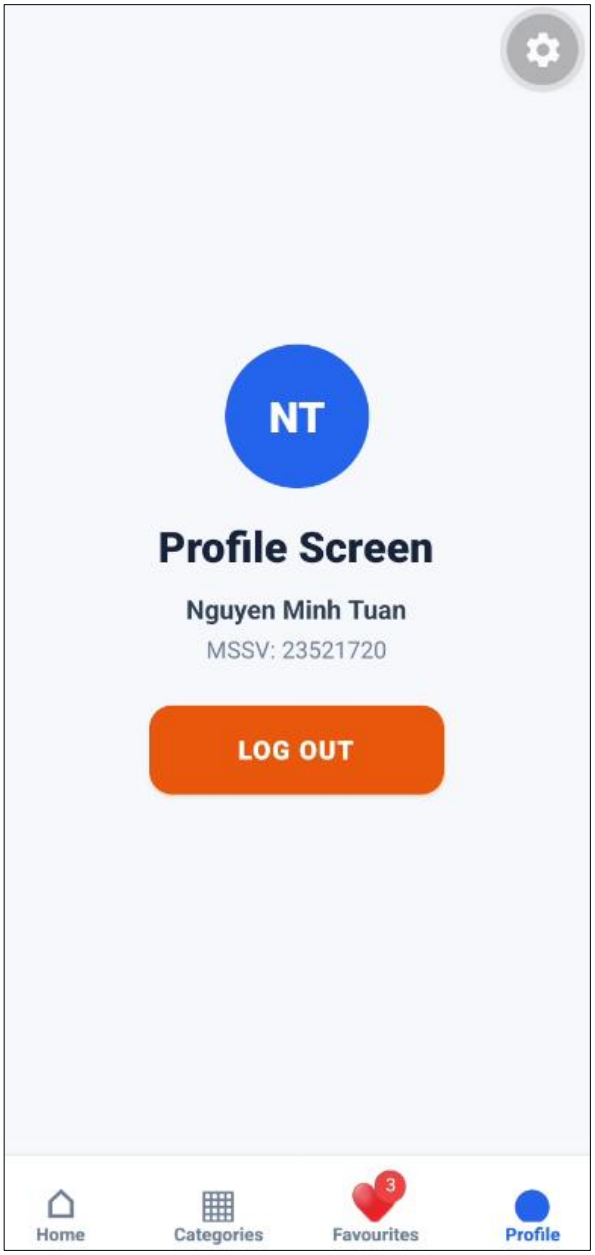





Welcome
Login failed
Incorrect email or password.
OK
LOG IN
New here? Sign up here!


Home Screen
Welcome to the main area of the app.
Home
Categories
3
Favourites
Profile


Favourites Screen
You currently have 3 favourite items.
3
Home
Categories
Favourites
Profile


NT
Profile Screen
Nguyen Minh Tuan
MSSV: 23521720
LOG OUT
3
Home
Categories
Favourites
Profile